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New TOR powered file sharing introduces features like public mode, anonymous Dropbox support and more. <https://digit.in/mar19-77>



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Samsung addressed the biggest complaint about Bixby button by allowing S10 users to choose what the button does. <https://digit.in/mar19-78>

AMD Radeon VII

Shooting for the stars

A surprise announcement at CES 2019, AMD unveiled to the world a flagship GPU in the form of the Radeon VII. Based on the Vega 20 GPU, the Radeon VII stood out because of the process shrink as well as the competitive pricing. Using TSMC's 7nm process, this is the world's first gaming graphics card on this process node and with that comes an improvement in performance that, AMD says, puts it right up in league of the RTX 2080. Priced at \$699 or ₹54,990, the Radeon VII supersedes the Vega64, but by exactly how much? That stands to be seen.

Specifications

Usually, when we have a new GPU within the same architecture (both are GCN5), the core count goes up and the clock speed scales to achieve the maximum possible value while not exceeding the thermal envelope. In the case of the Radeon VII, the core count actually went down. The Radeon VII has 60 Compute Units (CUs) whereas the Vega 64 has 64 CUs as the name suggests. However, AMD compensated for this by ramping up the base clock speed to 1400 MHz which is 153 MHz higher than the Vega 64. However, the TDP only went up to 300W which is a meagre 5W greater than the Vega64. Ideally, you'd see a much larger bump but we'd assume that the process shrink has helped with power savings which allowed for the overall improvement in performance.

What few people realised is that the core configuration seems exactly like that of the Radeon Instinct MI50, a AI/ML accelerator by AMD which was unveiled a few months prior. The price difference is shocking though. We've spotted the Radeon Instinct MI25 selling for \$4,900 online. The MI50 has much better specs and would have re-tailed for a lot more money. Obviously,



Price
₹54,990

manufacturing costs would be much lower than that but it begs the question as to how AMD managed to take a ridiculously expensive GPU and repackaged it for sale at a much lower price point. Multiple articles had been written around the launch time citing that the Bill of Materials (BOM) for the Radeon VII was too high for AMD to make any profit off of it. That's something we'll never know.

Build

The front and back plates are machined out of an aluminium alloy and most importantly, the sides are open. Older samples would have a single blower design so the Radeon VII has a marked difference. There's a massive copper vapour chamber connected to five heatpipes running along the length of the card. On top of this lies the heatsink assembly and the fans are seated in cutouts. The sheer size of the copper vapour chamber is jaw-dropping. For the I/O, we see 3x DVI and 1x HDMI ports.

Performance

In all our synthetic benchmarks, the Radeon VII inches past the RTX 2080 in Fire Strike Ultra but in Unigine we see the opposite. Then again, in our OpenCL benchmarks the Radeon VII maintains a good lead as Radeon cards have done for many generations in the past. In Fire Strike Ultra we managed to hit 6703 without any overclock.



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as the RTX 2080 in Shadow of the Tomb Raider but in Deus Ex Mankind Divided, the difference was about 9-12 FPS with the RTX 2080 in the lead. In Strange Brigade, the Radeon VII took the lead by about 6-8 FPS. Overall, the Radeon VII is evenly matched with the NVIDIA RTX 2080.

Verdict

Gaming benchmarks aside, AMD is marketing the massive HBM2 VRAM that the Radeon VII has. It obviously makes the Radeon more of a multi-purpose card making it good for content creation applications as well as gaming given how competitive it is with the NVIDIA RTX 2080. Though the card seems like a really good buy, you'll be hard pressed to find one in the market. In about a week since the February 7 launch, we've seen all the online listings from the card go out of stock. It's a pity because not only is the Radeon VII priced a lot lower than the RTX 2080 but with the video game bundle they've included, you're practically getting three AAA titles as well. Since you can't get your hands on one any time soon, all you can do now is wait for Navi cards due later this year.

—Mithun Mohandas

SPECIFICATIONS

STREAM PROCESSORS: 3840 | CU: 64 | BOOST CLOCK: 1750 MHz
| VRAM: 16 GB HBM2 | BUS-WIDTH: 4096-bit | CONNECTORS:
21x8PIN | TBP: <300W

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